Paper-3

ISRO

(Radiographer)

1. **Depression between the two tuberosities of humerus is known as**
   (a) Capitulam (b) Epicondyles
   (c) Bicipital groove (d) Trochlea

2. **Transverse plane is also called as**
   (a) Sagittal (b) Coronal
   (c) Axial (d) Median sagittal

3. **DEXA is used for**
   (a) CT scan
   (b) MRI
   (c) Bone mineral densitometry
   (d) Cardiac catheterisation

4. **In IVP, the first film after contrast injection is**
   (a) Nephrogram (b) Pyelogram
   (c) Ureterogram (d) Venogram

5. **Dental formula for permanent teeth is**
   (a) 2122/2122 (b) 2023/2023
   (c) 2103/2123 (d) 2123/2123

6. **IOPA is**
   (a) Intraoral paraapex
   (b) Intraoral periapical
   (c) Interoral para align
   (d) Interoral periapical

7. **The common carotid artery usually bifurcates at the level of _____ cervical vertebrae.**
   (a) 4th (b) 5th
   (c) 3rd (d) 2nd

8. **The functional tissue of lung is**
   (a) Alveoli
   (b) Bronchi
   (c) Respiratory bronchiole
   (d) Terminal bronchiole

9. **Elevation of the scapula is caused by all except**
   (a) Trapezius (b) Levator scapulae
   (c) Latissimus dorsi (d) Rhomboids major

10. **The number of cervical vertebrae**
    (a) 5 (b) 12
    (c) 7 (d) 8

11. **Protons are _____ charged particles.**
    (a) Positive (b) Negative
    (c) Neutral (d) None

12. **Ultrasound waves require _____ to travel.**
    (a) Medium (jelly, oil etc.)
    (b) Air
    (c) Vacuum
    (d) Gas

13. **Isotopes are those elements that have**
    (a) Same atomic number but different mass number
    (b) Different atomic number but same mass number
    (c) Equal number of protons and electrons
    (d) None of the above

14. **When X-ray beam falls on the body, they are**
    (a) Reflected (b) Absorbed
    (c) Transmitted (d) All of the above

15. **In modern X-ray tube, a vacuum is maintained equal to**
    (a) 10^-3 mm of Hg (b) 10^-2 mm of Hg
    (c) 10 mm of Hg (d) None

16. **Heat is dissipated in most of the X-ray tubes by**
    (a) Conduction, convection and radiation
    (b) Conduction and convection
    (c) Conduction and radiation
    (d) Convection and radiation

17. **Penetration of the X-ray is directly related to**
(a) KVP (b) mA
(c) Exposure time (d) Focus film distance

18. **The unit used to record the radiation received by radiation worker?**
(a) Roentgen (b) Rad
(c) Curie (d) Milli Sievert

19. **The normal frequency of ultrasound waves used in medical diagnostics range from**
(a) 20000 – 50000 Hz
(b) 50000<1 MHz
(c) 1MHz – 20 MHz
(d) >20 MHz

20. **Anode in rotating anode X-ray tube is made of**
(a) Tungsten – Nickel alloy
(b) Tungsten – Rhenium alloy
(c) Nickel – Tin alloy
(d) Tungsten – Aluminium alloy

21. **Compression in Mammography is important due to**
(a) Improved spatial resolution
(b) Improved contrast resolution
(c) Reduced patient dose
(d) All of the above

22. **Moseley's law relates to**
(a) Wavelength and angle of scattering
(b) Wavelength and intensity of X-ray
(c) Frequency and atomic number
(d) Frequency and applied voltage

23. **The usual thickness of the radiographic film is**
(a) 0.05 mm (b) 0.25 mm
(c) 0.75 mm (d) 1 mm

24. **Which is not true about intensifying screen?**
(a) Reduces patient dose
(b) Decreases spatial resolution
(c) Reduces blurring
(d) Increases loading of X-ray tube

25. **Ideal temperature of the dark room is**
(a) 10 – 15°C (b) 23 – 27°C
(c) 16 – 22°C (d) 28 – 32°C

26. **Intensifying screens are not used in**
(a) Mammography (b) Ultrasound
(c) Both (d) None

27. **Filter in the dark room used for the blue sensitive film is?**
(a) Amber (b) Gray
(c) Green (d) White

28. **The most commonly used photo sensitive agent?**
(a) Silver Chloride
(b) Silver Bromide
(c) Silver Iodide
(d) None of the above

29. **Fixing solutions used in radiography are**
(a) Alkaline (b) Acidic
(c) Neutral (d) None of the above

30. **In automatic processor, developer temperature is**
(a) 35 – 40°C (b) 50 – 55°C
(c) 18 – 20°C (d) 25 – 30°C

31. **The fracture of Scaphoid is best seen in**
(a) PA view
(b) Lateral view
(c) PA view with ulnar deviation
(d) PA view with radial deviation

32. **The bicipital groove of Humerus can be demonstrated in**
(a) Axial view of shoulder
(b) Tangential projection
(c) PA view of shoulder
(d) Lateral view

33. **The Launstein's projection view is basically done for**
(a) Pelvis
(b) Sacro Iliac joint
(c) Hip joint and upper third of Femur
(d) Chest region

34. **The specialised radiological examination of lungs and bronchial tree using an opaque contrast medium is called**
(a) Bronchoscopy (b) Bronchography
(c) Cystography (d) Urography

35. **To decrease the magnification of the heart in PA view of chest the film focal distance is kept at**
(a) 40 inches (b) 60 inches
(c) 72 inches (d) 36 inches

36. **The radiological examination of salivary glands and its ducts by means of contrast medium is termed as**
(a) Ductography (b) Sialography
(c) Sinography (d) Fistulogram

37. **Stenver projection view is basically done for**
(a) Base of skull
(b) Petrous portion of temporal bone
(c) Oblique view of mandible
(d) Orbit

38. **Patella is a**
(a) Flat bone (b) Long bone
(c) Irregular bone (d) Sesamoid bone

39. **Which view of the nasal bone shows bony nasal septum best?**
(a) Lateral projection
(b) AP projection
(c) Water's projection
(d) None of the above

40. **Skyline view is used for**
(a) Talus (b) Scapula
(c) Patella (d) Hip joint

41. **What position of the elbow best demonstrates the Olecranon process?**
(a) AP view (b) True lateral view
(c) Axial view (d) Oblique view

42. **What is the proper breathing instruction for an AP abdomen?**
(a) Full inspiration
(b) Rapid breathing
(c) Shallow breathing
(d) Full expiration

43. **Which of the following would best demonstrate fluid in the right pleural cavity?**
(a) Left lateral decubitus
(b) Right lateral decubitus
(c) Ventral decubitus
(d) Dorsal decubitus

44. **The patient's chin should be elevated during chest radiography to**
(a) Avoid superimposition on the apices
(b) Keep mid sagittal plane parallel
(c) Reduce patient motion
(d) Reduce patient dose

45. **The 10 days rule is applicable to**
(a) The first 10 days following the cessation of menstruation
(b) The first 10 days following the onset of menstruation
(c) The 10 days preceding the onset of menstruation
(d) 14 days before menstruation

46. **In India which of the following is the most significant in the development of radiation protection standard?**
(a) AERB (b) BARC
(c) NCRP (d) ICRP

47. **TLD is based on the phenomenon of**
(a) Induction
(b) Thermionic emission
(c) Thermoluminescence
(d) Photographic effect

48. **MCU is**
(a) Micturating Cysto Ureterography
(b) Modified Cysto Urethrogram
(c) Micturating Cysto Urethrography
(d) Micturating Cysto Urinography

49. **The primary factor that limits the maximum MA that can be used during a radiographic exposure is**
(a) Anode angle
(b) Focal spot size
(c) Cathode temperature
(d) Exposure time

50. **All the following are potential advantages of using a higher kV (90 rather than 70) in radiography except**
(a) Increased patient exposure
(b) Reduced X-ray tube heating
(c) Shorter exposure times
(d) Decreased area contrast

51. **In the adult, the spinal cord usually ends at the level of which vertebra?**
(a) L1 (b) S1
(c) L5 (d) S3

52. **The cochlea is located within the ______ bone.**
(a) Frontal bone (b) Occipital bone
(c) Temporal bone (d) Sphenoid bone

53. X-ray generators produce radiation through:
(a) Bremsstrahlung processes
(b) K-shell emission processes
(c) Both (a) and (b)
(d) Neither (a) or (b)

54. Increasing the magnetic field in MRI?
(a) Produces less susceptibility artifacts.
(b) Reduces the risk of tissue heating.
(c) Increases the signal to noise.
(d) Reduces the danger from metallic projectiles.

55. Newton's Inverse Square Law is useful in radiography because it indicates how the radiation intensity is affected by
(a) Radioactive decay
(b) Distance from the source
(c) The size of the source
(d) None of the above

56. As an ultrasound pulse moves through tissue in a patient's body it will undergo a change of all the following except:
(a) Frequency. (b) Amplitude (energy).
(c) Physical size. (d) Intensity.

57. The spatial resolution of an imaging system is most directly related to:
(a) Visibility of large, low contrast objects.
(b) Visibility of noisy images.
(c) Visibility of soft tissues.
(d) Visibility of anatomical detail.

58. The radio frequency energy used in MRI and X-ray have essentially the same:
(a) Velocity. (b) Photon energy.
(c) Wavelength. (d) Frequency.

59. The most appropriate instrument for measuring the scattered X-ray exposure from a patient is:
(a) Geiger counter
(b) Large ionization chamber
(c) Small ionization chamber
(d) Scintillation detector

60. The main component of radiographic noise is
(a) Structure mottle (b) Quantum mottle
(c) Random mottle (d) Graininess

61. Under the ionising radiation regulation, the annual dose limit for an adult worker is :
(a) 5 mSv (b) 50mSv
(c) 2mSv (d) 20mSv

62. Absorbed dose is measured in
(a) Sieverts (b) Man Sieverts
(c) Coulombs (d) Grays

63. One rad is equal to the absorption of
(a) 1 joules/kg (b) 0.01 Gy
(c) 100 joules/kg (d) 0.1 Gy

64. In the Photoelectric effect
(a) the gamma – ray is scattered
(b) all the gamma – ray energy is transferred to the photoelectron
(c) fluorescent X – ray emission never results
(d) none of the above

65. The intensity of the X-ray beam that leaves the X-ray tube is not uniform throughout all portions of the beam. This is called
(a) Heel effect
(b) Thermionic emission
(c) Stray radiation
(d) Tube rating

66. Why copper is used as an anode stem?
(a) High thermal conductivity
(b) Very hard
(c) Poor heat dissipation
(d) None of the above

67. Which is not a method for radiation protection?
(a) Dosimeter (b) Time
(c) Distance (d) Shielding

68. Which device is used to improve radiographic quality by reducing scattered radiation?
(a) X- ray table (b) Grids
(c) Film cassette (d) All of these

69. The X-ray tube is made up of
(a) Boro glass (b) Aluminium
(c) Pyrex glass (d) Beryllium

70. Fogging of film is caused by
(a) Pair production
(b) Pair annihilation
(c) Compton scattering
(d) Photo electric effect

71. The radiation warning symbol is a black ______ on a yellow background
(a) Trifoil (b) Dual foil
(c) Quad foil (d) Single foil

72. **The most damaging type of radiation is**
(a) Beta rays (b) X-rays
(c) Alpha rays (d) Gamma rays

73. **TLD should be worn at**
(a) The back side at the shoulder position
(b) Outside the lead apron stomach position
(c) Inside the lead apron the chest level
(d) Outside the lead apron collar position

74. **Who invented the cyclotron?**
(a) Lawrence (b) Roentgen
(c) Bergonie (d) Kerst

75. **The appropriate test to determine if an X-ray machine has adequate filtration is to measure the**
(a) Exposure output (b) HVL
(c) kV (d) Patient exposure

76. **The value of a CT number (in Hounsfield units) is determined primarily by**
(a) Matrix size (b) Slice thickness
(c) kV (d) Tissue density

77. **The efficiency of X-ray production (exposure/ heat unit) can generally be increased by increasing the**
(a) Focal spot size (b) kV
(c) mA (d) Exposure time

78. **Positron emission involves the ejection of:**
(a) an alpha particle
(b) a beta minus particle
(c) a beta plus particle
(d) a proton and a neutron

79. **In order to better visualize joint space in AP view of knee joint the tube is angled 5 to 7 degrees**
(a) Cranially
(b) Caudally
(c) Towards medial side
(d) Towards lateral side

80. **For AP view of coccyx the central ray is directed**
(a) 25° towards head
(b) 10° towards head
(c) 25° towards feet
(d) 10° towards feet

Answer and Explanation

1. **Ans. (c) Bicipital groove**
Exp. The depression between the two tuberosities of the humerus is known as the bicipital groove.

2. **Ans. (c) Axial**
Exp. The transverse plane is also called the axial plane.

3. **Ans. (c) Bone mineral densitometry**
Exp. DEXA (Dual-Energy X-ray Absorptiometry) is used for bone mineral densitometry.

4. **Ans. (a) Nephrogram**
Exp. In IVP (Intravenous Pyelogram), the first film after contrast injection is a nephrogram.

5. **Ans. (d) 2123/2123**
Exp. The dental formula for permanent teeth in humans is 2123/2123.

6. **Ans. (b) Intraoral periapical**
Exp. IOPA stands for Intraoral Periapical.

7. **Ans. (c) 3rd**
Exp. The common carotid artery usually bifurcates at the level of the 3rd cervical vertebra.

8. **Ans. (a) Alveoli**
Exp. The functional tissue of the lung is the alveoli.

9. **Ans. (c) Latissimus dorsi**
Exp. Elevation of the scapula is caused by the trapezius, levator scapulae, and rhomboids major, but not by the latissimus dorsi.

10. **Ans. (c) 7**
Exp. The number of cervical vertebrae is 7.

11. **Ans. (a) Positive**
Exp. Protons are positively charged particles.

12. **Ans. (a) Medium (jelly, oil etc.)**
Exp. Ultrasound waves require a medium (like jelly or oil) to travel.

13. **Ans. (a) Same atomic number but different mass number**
Exp. Isotopes are elements that have the same atomic number but different mass numbers.

14. **Ans. (d) All of the above**
Exp. When the X-ray beam falls on the body, it can be reflected, absorbed, or transmitted.

15. **Ans. (a) 10^-3 mm of Hg**
Exp. In modern X-ray tubes, a vacuum is maintained at approximately 10^-3 mm of Hg.

16. **Ans. (a) Conduction, convection and radiation**
Exp. Heat is dissipated in most X-ray tubes by conduction, convection, and radiation.

17. **Ans. (a) KVP**
Exp. Penetration of the X-ray tube is directly related to the kilovolt peak (KVP).

18. **Ans. (d) Milli Sievert**
Exp. The unit used to record the radiation received by a radiation worker is the milli Sievert.

19. **Ans. (c) 1MHz – 20 MHz**
Exp. The normal frequency range of ultrasound waves used in medical diagnostics is 1MHz – 20 MHz.

20. **Ans. (b) Tungsten – Rhenium alloy**
Exp. The anode in a rotating anode X-ray tube is made of a tungsten-rhenium alloy.

21. **Ans. (d) All of the above**
Exp. Compression in mammography improves spatial resolution, contrast resolution, and reduces patient dose.

22. **Ans. (c) Frequency and atomic number**
Exp. Moseley's law relates frequency and atomic number.

23. **Ans. (b) 0.25 mm**
Exp. The usual thickness of radiographic film is 0.25 mm.

24. **Ans. (c) Reduces blurring**
Exp. It is not true that intensifying screens reduce blurring.

25. **Ans. (c) 16 – 22°C**
Exp. The ideal temperature of the dark room is 16 – 22°C.

26. **Ans. (b) Ultrasound**
Exp. Intensifying screens are not used in ultrasound imaging.

27. **Ans. (a) Amber**
Exp. The filter used in the darkroom for blue sensitive film is amber.

28. **Ans. (b) Silver Bromide**
Exp. The most commonly used photosensitive agent in radiography is silver bromide.

29. **Ans. (b) Acidic**
Exp. Fixing solutions used in radiography are acidic.

30. **Ans. (a) 35 – 40°C**
Exp. In an automatic processor, the developer temperature is 35 – 40°C.

31. **Ans. (c) PA view with ulnar deviation**
Exp. The fracture of the scaphoid is best seen in the PA view with ulnar deviation.

32. **Ans. (a) Axial view of shoulder**
Exp. The bicipital groove of the humerus can be demonstrated in the axial view of the shoulder.

33. **Ans. (c) Hip joint and upper third of Femur**
Exp. The Launstein's projection view is done for the hip joint and upper third of the femur.

34. **Ans. (b) Bronchography**
Exp. The specialized radiological examination of the lungs and bronchial tree using an opaque contrast medium is called bronchography.

35. **Ans. (c) 72 inches**
Exp. To decrease the magnification of the heart in the PA view of the chest, the film focal distance is kept at 72 inches.

36. **Ans. (b) Sialography**
Exp. The radiological examination of salivary glands and their ducts by means of contrast medium is termed sialography.

37. **Ans. (b) Petrous portion of temporal bone**
Exp. Stenver projection view is done for the petrous portion of the temporal bone.

38. **Ans. (d) Sesamoid bone**
Exp. The patella is a sesamoid bone.

39. **Ans. (c) Water's projection**
Exp. Water's projection best shows the bony nasal septum.

40. **Ans. (c) Patella**
Exp. The skyline view is used for the patella.

41. **Ans. (b) True lateral view**
Exp. The true lateral view of the elbow best demonstrates the olecranon process.

42. **Ans. (d) Full expiration**
Exp. The proper breathing instruction for an AP abdomen is full expiration.

43. **Ans. (b) Right lateral decubitus**
Exp. The right lateral decubitus position would best demonstrate fluid in the right pleural cavity.

44. **Ans. (a) Avoid superimposition on the apices**
Exp. The patient's chin should be elevated during chest radiography to avoid superimposition on the apices.

45. **Ans. (b) The first 10 days following the onset of menstruation**

Exp. The 10 days rule is applicable to the first 10 days following the onset of menstruation.

46. **Ans. (a) AERB**

Exp. In India, the Atomic Energy Regulatory Board (AER (b) is most significant in the development of radiation protection standards.

47. **Ans. (c) Thermoluminescence**

Exp. TLD (Thermoluminescent Dosimeter) is based on the phenomenon of thermoluminescence.

48. **Ans. (c) Micturating Cysto Urethrography**

Exp. MCU stands for Micturating Cysto Urethrography.

49. **Ans. (b) Focal spot size**

Exp. The primary factor that limits the maximum mA that can be used during a radiographic exposure is the focal spot size.

50. **Ans. (a) Increased patient exposure**

Exp. Using a higher kV in radiography can lead to increased patient exposure.

51. **Ans. (a) L1**

Exp. In adults, the spinal cord usually ends at the level of the L1 vertebra.

52. **Ans. (c) Temporal bone**

Exp. The cochlea is located within the temporal bone.

53. **Ans. (c) Both (a) and (b)**

Exp. X-ray generators produce radiation through both Bremsstrahlung processes and K-shell emission processes.

54. **Ans. (c) Increases the signal to noise**

Exp. Increasing the magnetic field in MRI increases the signal-to-noise ratio.

55. **Ans. (b) Distance from the source**

Exp. Newton's Inverse Square Law is useful in radiography because it indicates how the radiation intensity is affected by the distance from the source.

56. **Ans. (a) Frequency**

Exp. As an ultrasound pulse moves through tissue in a patient's body, its frequency does not change.

57. **Ans. (d) Visibility of anatomical detail**

Exp. The spatial resolution of an imaging system is most directly related to the visibility of anatomical detail.

58. **Ans. (a) Velocity**

Exp. The radio frequency energy used in MRI and X-ray have essentially the same velocity.

59. **Ans. (b) Large ionization chamber**

Exp. The most appropriate instrument for measuring the scattered X-ray exposure from a patient is a large ionization chamber.

60. **Ans. (b) Quantum mottle**

Exp. The main component of radiographic noise is quantum mottle.

61. **Ans. (d) 20 mSv**

Exp. Under the ionizing radiation regulation, the annual dose limit for an adult worker is 20 mSv.

62. **Ans. (d) Grays**

Exp. Absorbed dose is measured in grays.

63. **Ans. (b) 0.01 Gy**

Exp. One rad is equal to the absorption of 0.01 Gy.

64. **Ans. (b) All the gamma-ray energy is transferred to the photoelectron**

Exp. In the photoelectric effect, all the gamma-ray energy is transferred to the photoelectron.

65. **Ans. (a) Heel effect**

Exp. The intensity of the X-ray beam that leaves the X-ray tube is not uniform throughout all portions of the beam, known as the heel effect.

66. **Ans. (a) High thermal conductivity**

Exp. Copper is used as an anode stem due to its high thermal conductivity.

67. **Ans. (a) Dosimeter**

Exp. A dosimeter is not a method for radiation protection but a device to measure exposure.

68. **Ans. (b) Grids**

Exp. Grids are used to improve radiographic quality by reducing scattered radiation.

69. **Ans. (c) Pyrex glass**

Exp. The X-ray tube is made up of Pyrex glass.

70. **Ans. (c) Compton scattering**

Exp. Fogging of film is caused by Compton scattering.

71. **Ans. (a) Trifoil**

Exp. The radiation warning symbol is a black trifoil on a yellow background.

72. **Ans. (c) Alpha rays**

Exp. Alpha rays are the most damaging type of radiation.

73. **Ans. (c) Inside the lead apron the chest level**

74. **Ans. (a) Lawrence**

Exp. The cyclotron was invented by Lawrence.

75. **Ans. (b) HVL**

Exp. The appropriate test to determine if an X-ray machine has adequate filtration is to measure the HVL (Half-Value Layer).

76. **Ans. (d) Tissue density**

Exp. The value of a CT number (in Hounsfield units) is determined primarily by tissue density.

77. **Ans. (b) kV**

Exp. The efficiency of X-ray production can generally be increased by increasing the kV.

78. **Ans. (c) A beta plus particle**

Exp. Positron emission involves the ejection of a beta plus particle.

79. **Ans. (b) Caudally**

Exp. To better visualize the joint space in the AP view of the knee joint, the tube is angled 5 to 7 degrees caudally.

80. **Ans. (d) 10° towards feet**

Exp. For the AP view of the coccyx, the central ray is directed 10° towards the feet.

□□□□